

Question			Marking details	Marks available																																																			
				AO1	AO2	AO3	Total	Maths	Prac																																														
5	(a)	(i)	Absorb alpha and beta or only allow gamma through. Accept absorb beta only [due to 40 cm gap]			1	1		1																																														
		(ii)	Background radiation (1) To check for consistency / check for variations (1) Don't accept for a mean			2	2		2																																														
	(b)		<table><tr><th rowspan="2">Distance, r / m</th><th colspan="4">Count</th><th rowspan="2">Uncertainty in mean count</th></tr><tr><th>First reading</th><th>Second reading</th><th>Third reading</th><th>Mean count</th></tr><tr><td>Without source</td><td>19</td><td>22</td><td>25</td><td>22</td><td>3</td></tr><tr><td>1.000</td><td>110</td><td>130</td><td>136</td><td>125</td><td>13</td></tr><tr><td>0.800</td><td>195</td><td>165</td><td>178</td><td>179</td><td>15 or 14 or 16</td></tr><tr><td>0.600</td><td>270</td><td>316</td><td>300</td><td>295</td><td>23 or 21 or 22 or 24 or 25</td></tr><tr><td>0.400</td><td>661</td><td>604</td><td>651</td><td>639</td><td>29</td></tr><tr><td>Without source</td><td>20</td><td>20</td><td>25</td><td>22</td><td>3</td></tr></table> 1 mark per column.	Distance, r / m	Count				Uncertainty in mean count	First reading	Second reading	Third reading	Mean count	Without source	19	22	25	22	3	1.000	110	130	136	125	13	0.800	195	165	178	179	15 or 14 or 16	0.600	270	316	300	295	23 or 21 or 22 or 24 or 25	0.400	661	604	651	639	29	Without source	20	20	25	22	3		2		2	2	2
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